Team 05 CIT433027

Diffusion Policies for 19-DoF Grasping

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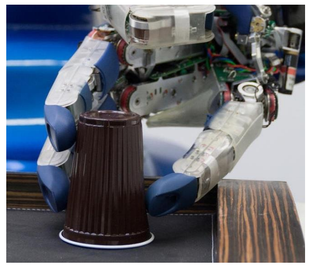
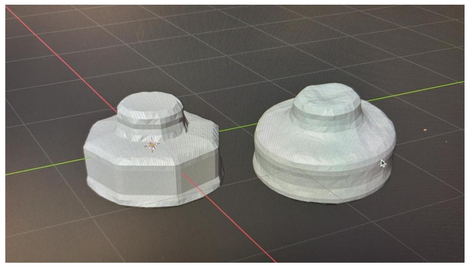
Problem & Motivation

- Existing grasping research primarily targets **6-DoF parallel-jaw grippers**
- **High-DOF multi-fingered hands** induce a large, highly multi-modal grasp space
- Traditional planning methods struggle in **high-dimensional configuration spaces**
- **Diffusion models** enable generative modelling of complex high dimensional grasp synthesis

Dexterous Robotic Hand & Voxel representation

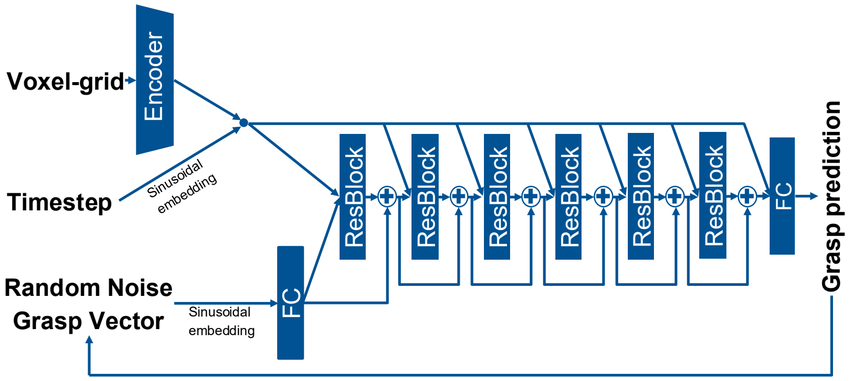
- **19 DoF Hand:** 7-DoF wrist pose plus 12-DoF finger joint angles
- **3D Geometry:** Objects are transformed into a dense Signed Distance Field (SDF) at a 32^3 resolution.





System Architecture

- **Spatial Encoding:** 3D-CNN processes the object to extract a dense geometric feature vector
- **State Expansion:** The diffusion timestep and the initial 19D noise vector are projected into high-dimensional space
- **Denoising Backbone:** A deep Residual MLP processes the expanded noise iteratively using the visual-temporal context
- **Output:** The final synthesized 19-DoF dexterous grasp configuraltion



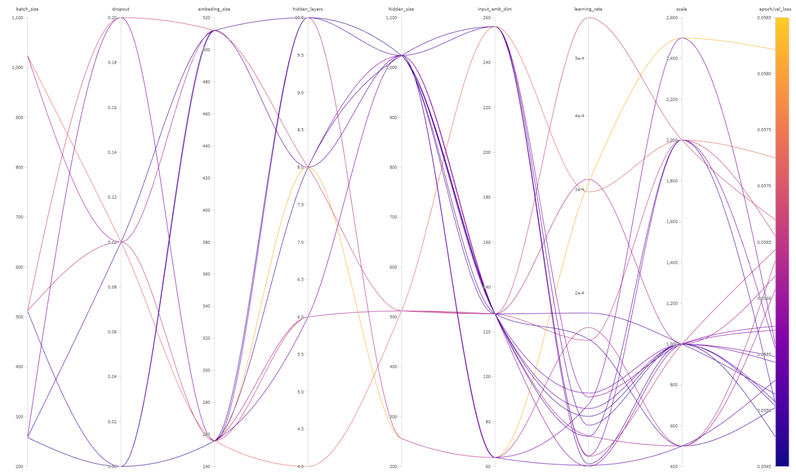
Learning & Conditioning Mechanism

- **High-Frequency Mapping:** Sinusoidal embeddings expand the input, allowing the network to learn fine-grained 19-DoF finger movements
- **AdaLN Integration:** AdaLN uses the voxel and time data to calculate scale and shift parameters, dynamically adjusting how strongly the 3D object guides the network at each layer

$$\mathbf{x_{out}} = \mathbf{LayerNorm}(\mathbf{x_{in}}) \cdot (\mathbf{1} + \mathbf{scale}) + \mathbf{shift}$$

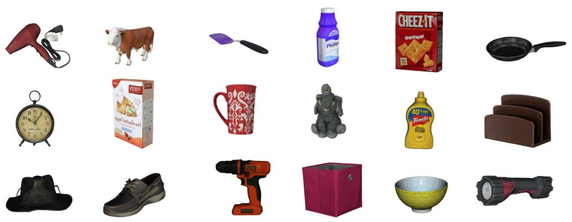
Hyperparameter Search

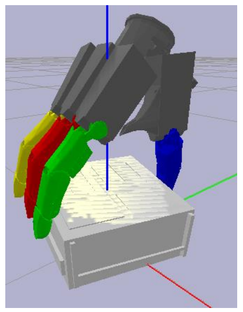
- **Diminishing Returns:** Hidden size is the primary performance driver. However the gain is marginal for a 200% parameter increase
- **Architectural Robustness:** The Model exhibits high stability maintaining strict validation loss bounds ($\mathrm{MSE} \sim 0.055,\ \sigma < 0.001$)



Simulation

- **Dataset:** MultiGripperGrasp (329 household objects)
- **Physics Engine:** PyBullet with impedance controller
- **Evaluation:** Object stability – remains in hand



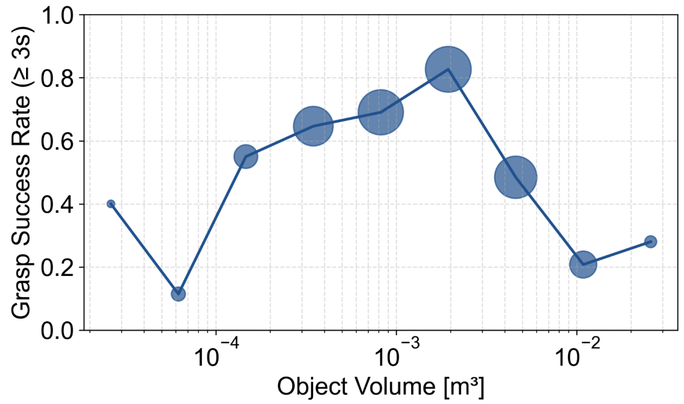


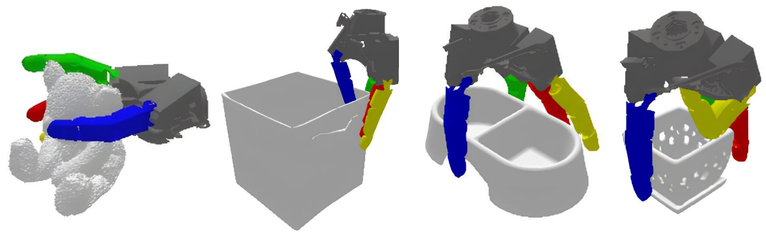
Results

- **Feasable grasp generation** achieved for all test obejcts
- **Stable grasps** for the majority of objects across trials
- Performances object size dependent

Success and Failure Rates

Grasp Success Rate (%)				Grasp Failure Rate (%)	Object Success Rate (%)	Total Grasps
> 3s	> 2s	> 1s	≥0s			
60.48	2.00	3.04	34.47	39.52	80.85	1654





Conclusion

- Diffusion policy captures multi-modal 19-DoF grasps
- Voxel representation enables generalization to unseen objects
- Simulation ensures reliable grasp evaluation
- Hyperparameter tuning gives negligeable improvement in performance

Future

- **Sparse 3D Encoders:** Scaling from dense 32^3 grids to efficient, high-resolution SDFs to capture intricate surface details
- **Sim2Real Deployment:** Validating the generated 19-DoF grasp configurations on a real robotic hand